BATCH3_RTA_GEOM

PREREG: BATCH3_RTA_GEOM (Map-C12 to Map-C15; status of C6/C10/C11/E1–E3)

Written 2026-10-24 at about 7:05 PM CDT, before any BATCH3 test. Owner: Arvin Hampton. New files only. CORE_FREEZE. All maps are **Category B**; a pass is a survived prediction, not a bridge. No Π, grading or T3 q-series is invented. 67/68 are not targets. Runs: nice, OMP_NUM_THREADS=1, ulimit -v 3000000. Order19 (affine A2c) is not touched.

Written residual objects used (sources quoted in reports/crypto/RPP_spec_v1/RESIDUAL_T3_predicate_sources.md §2)

- O_res = the 539 least labels of the charge class C_{q⋆} in Ω′ after clearing f_max = 21 (Object539_NonCircular_Construction §3.5). Canonical core: x_j = 21 + 9j, j = 0..538 (as reproduced there: x = 21+9j).
- Fibres: β♯(x_j) = j mod 8 on the sorted index (Discrete_Patterns_Residual_Carrier §0).
- Towers: Ω = {0..4879} cut into 243 consecutive blocks, the first 20 of size 21 and the rest of size 20; τ(x) = block index. The written count is 353 same-tower adjacent pairs.
- **Search result (task 1):** no written 539-step Π, no ±1 grading on O_res and no path predicate exist (see NOTES for the paths searched). C6, C10, C11 and E1–E3 stay OPEN and are not re-hashed.

Map-C12: orthant-angle norm form → equidistribution over the fibres

Source: ε = 6+√35 (orthant angle ½ log ε), with norm form N(a,b) = a² − 35b². Target: O_res with β♯. Prediction (NEW): let S = {x ∈ O_res : x = |a² − 35b²| for some integers a, b}. The counts of S in the 8 β♯-fibres differ by at most 1 (max − min ≤ 1), and S ≠ ∅. PASS iff both. (Search bound |b| ≤ 200, which exceeds the reduction bound √(x·ε/35) ≈ 41.)

Map-C13: RG1′ reflection → carrier reversal

Source: RG1′ (zeros iff ρ_m or ρ_u preserves the companion; an endpoint reflection). Target: sorted core with towers τ. Prediction (NEW): the reversal σ(j) = 538 − j maps the set of same-tower adjacent pairs {(j, j+1) : τ(x_j) = τ(x_{j+1})} onto itself. PASS iff σ preserves the set exactly. Also required: the reproduced pair count equals the written 353 (a gate; if the gate fails, the result is VOID, not FAIL).

Map-C14: law-vector form vs norm form (two classes of disc 140) → O_res

Source: $2Q(n) = 2n_1^2 + 14n_1n_2 + 7n_2^2$ (discriminant 140, the 35-component law vector form) vs $a^2 - 35b^2$ (disc 140). Target: O_res. Prediction (NEW): $S_Q = \{x \in O_{res} : x = |2a^2+14ab+7b^2|\}$ and S (from C12) are disjoint, and S_Q is non-empty. PASS iff both. (Search $|a|, |b| \le 300$.)

Map-C15: order19 A2c corner → fibre count (feeds from order19). OPEN

Prediction: let k = the number of the 105 companions with P-even corner A2c ≠ 0 (order19 exact output). Then k ≡ 0 (mod 8), i.e. it is compatible with an equivariant spread over the 8 β♯-fibres. PASS iff 8 | k. Next measurement: read k from order19's final data once it is complete. Do not compute it here. # RESULTS: BATCH3_RTA_GEOM PREREG.md sha256 b57478f74b920a6a059391ea2d937a6f1659b824baa2a8a2e32bdcae8aba21b2, hashed before tests. Script scripts/b3.py (hash in scripts/SHA256SUMS); data data/b3.json; log logs/b3.log. Exact integer arithmetic, under 1 s.

Map	Prediction	Measured	Verdict	Cat
C12	norm-form set $S \subset O_{res}$ is non-empty and equidistributed over β♯	$S = \varnothing$	**FAIL**	B
C13	reversal σ preserves same-tower adjacent pairs; gate: count = 353	reproduced 297 adjacent pairs (≠ 353)	**VOID** (gate failed, by rule)	B
C14	S_Q non-empty and disjoint from S	$S_Q = \varnothing$	**FAIL**	B
C15	8 | k (order19 A2c nonzero count)	awaits order19	**OPEN**	B

Map	Prediction	Measured	Verdict	Cat
C6, C10, C11, E1–E3	—	no written Π, grading or predicate found	**OPEN**	B

Structural reasons (Category A, elementary, found after the run; verdicts unchanged)

- C12/C14: every x ∈ O_res satisfies 3 | x and 9 ∤ x (x = 21+9j ≡ 3 mod 9). Mod 3: $a^2 - 35b^2 \equiv a^2 + b^2$ and $2a^2+14ab+7b^2 \equiv a^2 + (a+b)^2$. Both vanish mod 3 only when 3 | a and 3 | b, which forces 9 | value. So no x ∈ O_res is represented by either form (Category A). The maps were badly posed, and this is not a near-miss.
- C13: the written “353 same-tower pairs” is Σ_t C(n_t, 2) (all unordered pairs within a tower). Recomputed post-hoc: 353 exactly. Adjacent pairs number 538 − Σδf = 538 − 241 = 297. The gate used the wrong reading, so the verdict is VOID. A re-test would need a new prereg. # NOTES: BATCH3_RTA_GEOM ## Task 1: search for a written Π / grading / fibre labelling / predicate Searched (rg, case-insensitive) /workspace/quantum-comp/** and /tmp/repo539/** for: “539-step (word|path|trajectory|sequence)”, “Π =|=|resonant path problem”, “grading on O_res| ±1 grading|sign grading|Z/2 grading”, “block label|fibre block [0-9]|block index”, “residual predicate|path predicate”. Hits read:
- reports/crypto/RPP_formal_statement/RPP_formal_statement.md: Π is defined as the fixed 539-step T3 body of HQH-539 on u_0 = SHA3-512(m‖s), with “optional residual constraints” that are never specified.
- reports/crypto/RPP_spec_v1/RESIDUAL_T3_predicate_sources.md (2026-10-23): explicit finding that **no source defines a map from a path or T3 value to O_res, a fibre or the permanent class**. It also quotes Architecture_A_Ores_Programme §6, which bars carrier functions on T♯-orbits.
- reports/crypto/RPP_spec_v1/scripts/n_residual_filter.py: a *candidate* fibre map (u mod 539 → interval blocks), labelled as a candidate, not sourced.
- U1_LEDGER_PPT_R_Obstruction_Written_Data_2026-09-14.md (both trees): PR9, “A count is not a clock.” **Found:** written definitions of the fibre labelling β♯(x_j) = j mod 8 (residue classes of the sorted index, NOT intervals), the canonical core x_j = 21+9j and towers τ. **Not found:** any 539-step Π word, any ±1 grading on O_res, any path predicate. C6, C10, C11 and E1–E3 stay OPEN and were not re-hashed.

Correction to batch 1

MAPS_CANDIDATES Map-C4 used consecutive interval blocks. The written fibres are j mod 8. C4’s FAIL stands for the interpretation it preregistered, but that interpretation does not match the source. A source-faithful C4 would need a new prereg.

Other notes

- 67/68 were not used as targets.
- Order19 was not touched. C15 is written so that order19’s final k can be plugged in.

- **Sharpest next lever:** the charge-class obstruction (x ≡ 3 mod 9) shows that any modular→O_res map through quadratic-form representation must respect the mod-9 charge. The next well-posed candidate should use a written carrier function that is mod-9-blind: β♯, τ, or the permanent class on K_9 × P_{B′}. Example: RG1′-type reflections acting on the product complex K_9 × P_{B′}, predicting the sign of α(a,b) = sgn(a−b) under σ. This is written and Category A on the residual side.
- CORE_FREEZE; new files only; no push, no sends.